📘 Data Types & Built-in Functions

1. Data Types

In Python, everything is an object, and every object has a data type. Data types are essentially classes, and variables are instances (objects) of those classes.

A. Number

Used to store numeric values. They are **immutable**.

1. Integer (`int`)

- **Description:** Represents whole numbers, positive or negative, without a decimal point.
- **Examples:** `10` , `-5` , `0` , `12345` .
- **Note:** Integers in Python 3 have unlimited length (only limited by system memory).

2. Floating Point (`float`)

- **Description:** Represents real numbers with a decimal point. They can also be represented in scientific notation (e.g., `1.2e3`).
- **Examples:** `3.14` , `-0.001` , `2.0` , `1.5e2` (which is 150.0).
- **Note:** Floating-point numbers are approximate values due to the way they are stored in memory.

3. Complex (`complex`)

- **Description:** Represents complex numbers with a real and an imaginary part. The imaginary part is denoted by `j` or `J` .
- **Examples:** `3 + 4j` , `5j` , `2 - 1j` .
- **Note:** Access real part via `z.real` and imaginary part via `z.imag` (where `z` is the complex variable).

B. Boolean (`bool`)

- **Description:** Represents one of two truth values: `True` or `False` . It is a subclass of `int` , where `True` acts like 1 and `False` acts like 0.
- **Usage:** Used for logical operations and conditional statements.
- **Examples:** `is_registered = True` , `is_valid = 3 > 2` (evaluates to `True`).

C. Sequence

A sequence is an ordered collection of items, where each item can be accessed via its index.

1. String (`str`)

- **Description:** Represents a sequence of characters (alphanumeric, symbols, etc.) enclosed within single, double, or triple quotes.
- **Examples:** `'Hello'`, `"CS Class"`, `'''Multi-line String'''`.
- **Nature:** IMMUTABLE (cannot be changed after creation).

2. List (`list`)

- **Description:** Represents an ordered, mutable collection of items. Items can be of different data types. Defined using square brackets `[]`.
- **Examples:** `[1, 2, 3]`, `['a', True, 10.5]`, `[]` (empty list).
- **Nature:** MUTABLE (can be changed after creation – add, remove, or modify elements).

3. Tuple (`tuple`)

- **Description:** Represents an ordered, immutable collection of items. Defined using parentheses `()`.
- **Examples:** `(1, 2, 3)`, `('a', True, 10.5)`, `()` (empty tuple).
- **Nature:** IMMUTABLE (cannot be changed after creation).

D. None (`NoneType`)

- **Description:** Represents the absence of a value or a null value. It is not the same as `0`, `False`, or an empty string.
- **Usage:** Often used to initialize a variable or to signify that a function has no return value.
- **Example:** `result = None`

E. Mapping (`dict` – Dictionary)

- **Description:** Represents an unordered, mutable collection of key-value pairs. Keys must be unique and immutable (e.g., strings, numbers, tuples), while values can be of any data type and are accessed via their key. Defined using curly braces `{}`.
- **Examples:** `{'name': 'Alice', 'age': 25}`, `{}` (empty dictionary).
- **Nature:** MUTABLE (can be changed after creation – add, remove, or update key-value pairs).

F. Mutable vs Immutable Data Types

This is a crucial concept in Python.

Feature	MUTABLE	IMMUTABLE
Definition	Objects whose state (value) can be changed after creation.	Objects whose state (value) **cannot** be changed after creation.
Memory	When modified, the object remains in the same memory location, and its content is updated.	When modified, a new object is created in a new memory location.
Data Types	`list`, `dict`, `set` (not in syllabus but good to know)	`int`, `float`, `complex`, `bool`, `str`, `tuple`

G. Dynamic Typing

Python is a **dynamically typed** language. This is one of its most important and beginner-friendly features.

What is Dynamic Typing?

- In dynamically typed languages, the **data type of a variable is determined at runtime** (during program execution), not at compile time.
- You **do not need to declare** the data type of a variable before using it. The interpreter infers the type based on the value assigned to it.
- The **same variable can hold values of different data types** at different points in the program.

Key Characteristics

- **No explicit type declaration:** Unlike languages like C, C++, or Java, you don't write `int x = 5`; you simply write `x = 5`.
- **Type is associated with the object, not the variable:** The variable is just a name (or reference) that points to an object in memory. The object itself knows its type.
- **Type can change:** A variable can be reassigned to a completely different data type.

```
# Variable 'x' starts as an integer

x = 10

print(x, type(x))  # Output: 10 <class 'int'>
```

```
# Same variable 'x' is now assigned a string
x = "Hello"
print(x, type(x))  # Output: Hello <class 'str'>

# Now 'x' becomes a floating-point number
x = 3.14
print(x, type(x))  # Output: 3.14 <class 'float'>

# Now 'x' becomes a boolean
x = True
print(x, type(x))  # Output: True <class 'bool'>

# Now 'x' becomes a list
x = [1, 2, 3]
print(x, type(x))  # Output: [1, 2, 3] <class 'list'>
```

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2. Common Built-in Functions

Built-in functions are pre-defined functions in Python that are always available for use.

1. `eval()`

- **Purpose:** Evaluates a string expression and returns the result. It effectively treats the string as a Python expression and executes it.
- **Parameter(s):**
  - `source` : (Compulsory) The string expression to be evaluated.
- **Description:** Parses the string argument and compiles it to Python bytecode, which is then executed.

```
# Example 1: convert string input
value = eval(input("Enter a value"))
```

```
print(value)  # Output: 57.12 -- if the input is 57.12 float value
print(type(value))  # Output: <class 'float'>

# Example 2: arithmetic expression
x = 10
result = eval("x * 2")  # Evaluates 10 * 2
print(result)  # Output: 20
```

2. `type()`

- **Purpose:** Returns the data type (class) of the argument passed to it.
- **Parameter(s):**
  - `object` : (Compulsory) The object whose type you want to know.
- **Description:** Returns a type object that tells you the class of the given object.

```
a = 10
b = 3.14
c = "Hello"
d = [1, 2]
e = (1, 2)
f = {'key': 'value'}

print(type(a))  # Output: <class 'int'>
print(type(b))  # Output: <class 'float'>
print(type(c))  # Output: <class 'str'>
print(type(d))  # Output: <class 'list'>
print(type(e))  # Output: <class 'tuple'>
print(type(f))  # Output: <class 'dict'>
```

3. `id()`

- **Purpose:** Returns the unique identifier (memory address) of an object.
- **Parameter(s):**
  - `object` : (Compulsory) The object whose memory address you need.
- **Description:** Every object in Python is stored at a specific memory location. The `id()` function returns this address as an integer. This is helpful to understand the concept of mutability and how Python handles variables.

```
x = 10
print(id(x))  # Output: Some memory address like 140735143830864

y = x  # y now points to the same memory location as x
print(id(y))  # Output: Same memory address as x

list1 = [1, 2]
print(id(list1))  # Output: e.g., 140281465641536

list2 = list1
print(id(list2))  # Output: Same as list1's address
```

4. `chr()`

- **Purpose:** Returns the character corresponding to a given Unicode code point (integer).
- **Parameter(s):**
  - `i` : (Compulsory) An integer (within the valid Unicode range, 0 to 1,114,111).
- **Description:** Converts an integer to its character representation. For example, 65 maps to 'A'.

```
print(chr(65))  # Output: A
print(chr(97))  # Output: a
print(chr(48))  # Output: 0 (zero)
```

```
print(chr(8364))  # Output: € (Euro symbol)
```

5. `ord()`

- **Purpose:** Returns the Unicode code point (integer) for a given character. It is the inverse of `chr()`.
- **Parameter(s):**
  - `c`: (Compulsory) A string of length one (a single character).
- **Description:** Converts a character to its integer representation.

```
print(ord('A'))  # Output: 65

print(ord('a'))  # Output: 97

print(ord('0'))  # Output: 48

print(ord('€'))  # Output: 8364
```

6. `len()`

- **Purpose:** Returns the number of items (length) of an object. The object must be a sequence type (string, list, tuple) or a collection (dictionary).
- **Parameter(s):**
  - `s`: (Compulsory) The sequence or collection object.
- **Description:** Counts the number of elements.
  - For strings: Counts the number of characters.
  - For lists/tuples: Counts the number of elements.
  - For dictionaries: Counts the number of key-value pairs.

```
text = "Computer Science"

print(len(text))  # Output: 16 (Spaces count too!)


my_dict = {'name': 'John', 'age': 30, 'city': 'NYC'}

print(len(my_dict))  # Output: 3 (Number of keys)
```

7. `sum()`

- **Purpose:** Adds all the items in an iterable (like a list or tuple) and returns the total sum.
- **Parameter(s):**
  - `iterable` : (Compulsory) A sequence of numbers (list, tuple, etc.) to be summed.
- **Description:** Calculates the sum of all elements in the iterable. The elements must be numeric (int, float).

```
numbers = [10, 20, 30, 40]

total = sum(numbers)

print(total)  # Output: 100


# With start parameter

total_with_start = sum(numbers, 5)

print(total_with_start)  # Output: 105 (100 + 5)


# Sum of float values

float_nums = [1.5, 2.7, 3.2]

print(sum(float_nums))  # Output: 7.4
```

8. `min()`

- **Purpose:** Returns the smallest item from an iterable or from two or more arguments.
- **Parameter(s):**
  - `iterable` : (Compulsory) a single iterable.
- **Description:** Finds the minimum value. Works with numbers, strings (lexicographic comparison), and other comparable data types.

```
# With iterable

numbers = [45, 12, 78, 23, 9, 56]

print(min(numbers))  # Output: 9
```

```
# With multiple arguments
print(min(100, 200, 50, 75, 150))  # Output: 50

# With strings (alphabetical order)
words = ["apple", "banana", "cherry", "date"]
print(min(words))  # Output: apple (smallest alphabetically)
```

9. `max()`

- **Purpose:** Returns the largest item from an iterable or from two or more arguments.
- **Parameter(s):**
  - `iterable` : (Compulsory) a single iterable.
- **Description:** Finds the maximum value. Works with numbers, strings (lexicographic comparison), and other comparable data types.

```
# With iterable
numbers = [45, 12, 78, 23, 9, 56]
print(max(numbers))  # Output: 78

# With multiple arguments
print(max(100, 200, 50, 75, 150))  # Output: 200

# With strings (alphabetical order)
words = ["apple", "banana", "cherry", "date"]
print(max(words))  # Output: date (largest alphabetically)

# With mixed numeric types
mixed = [5, 10.5, 3, 8.2, 15]
print(max(mixed))  # Output: 15
```